

國立臺灣大學 學期 考試答案卷

National Taiwan University Midterm/Final Examination Answer Sheet

記	分	教師簽名或蓋章
Score		Lecturer's signature

課程編號

Course no. _____

科目

Course title _____

考試日期 年 月 日

Date Y M D

從此處開始寫起。試卷用紙務須節用。非經主試認可不得續用其他紙張作答。

Please write from here.

學院 管理

College

學系

Department

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Division

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Year

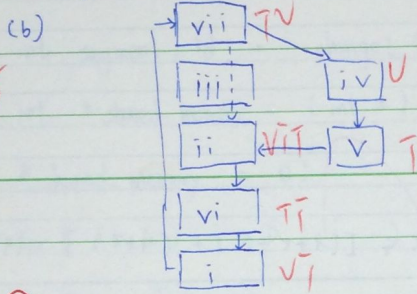
學號

Student ID no. BD3705042

姓名

許承佑

1. (a) Device controllers read or write data stored in device. Device drivers tell operating system the methods to communicate with device. *device controller*



and code can be written faster

2. It is easy for human to read and understand the codes with higher-level language. Programmers can share codes with teammates, debug with compilers and debuggers, and develop more modules.

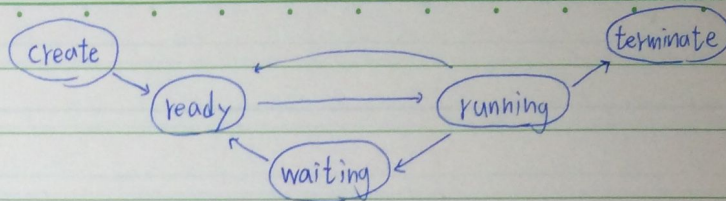
② OS written in high-level language is easier to port to some other hardware.

3. Most commands in shell are to implement file system. There are some commands frequently used by users. If a command interpreter contains all the commands, the size of interpreter will be huge but quick. Some commands are rare to use and the interpreter loaded in memory will be waste. The other way to implement the commands through system program is different. In this case, interpreter can be small in size but take a long time to execute a command.

4. (a) Invoking a system call is a tedious work with few things to change. APIs provide a series of functions to involve system call. First, software interrupt will make CPU change to kernel mode to handle the emergency. There is a system call number representing a system program that CPU will execute in kernel space. When the event is done, CPU will back to user mode and resume the program counter to continue the process.

(b) It is tough to invoke system calls directly. With the help of API functions, programmers can easily complete the task without concerning about the methods to implement system calls.

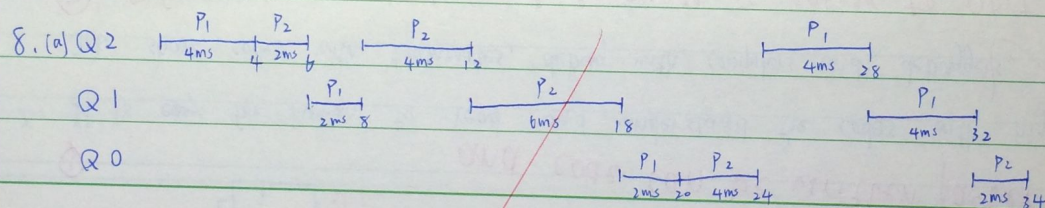
5. (a) ready (e) ready
 (b) ready (f) ready
 (c) running (g) ready
 (d) waiting



6(a) Every process contains program, stack, heap and more in memory. If two or more processes need to use the same data, the data will be copied to other process' stack or heap and cost more memory spaces. Different threads can share memory with each other without more memory spaces.

(b) The many-to-~~many~~^{one} model has a kernel thread allocates many user thread. It can be quick in executing because it will not involve system call. The disadvantage of many-to-~~many~~^{one} model is that if one user thread is blocked, all the threads will be blocked, too. The one-to-one model can fix that problem but is limited in number.

7. When there are infinite processes with high priority, it is possible that process with lower priority will never be executed. There is a number increasing all the time called "time". If the age of process is old enough, the priority will rise until it is on the top.



(b) $[(2+10) + (2+2+8)] \div 2 = 12 \text{ (ms)}$

9,

```
int main (int argc, char *argv[]) {  
    int x = atoi(argv[1]);  
    int y = atoi(argv[2]);  
    int i;  
  
    pid_t pid;  
    const int sh_size = 4;  
    int segment_id = shmget (IPC_PRIVATE, sh_size, S_IRUSR | S_IWUSR);  
    int *shared_memory = (int *) shmat (segment_id, NULL, 0);  
    *shared_memory = 0;  
  
    for (i = 2; i > 0; i--) {  
        pid = fork();  
        if (pid < 0) {  
            fprintf(stderr, "Fork Failed");  
            exit(-1);  
        }  
        else if (pid == 0) {  
            if (i == 2)  
                *shared_memory += 2 * x;  
            if (i == 1)  
                *shared_memory += 3 * y;  
            shmdt (shared_memory);  
            exit(0);  
        }  
        else {  
            wait();  
        }  
    }  
  
    *shared_memory -= 5;  
    printf ("the result is %d\n", *shared_memory);  
    shmdt (shared_memory);  
    shmctl (segment_id, IPC_RMID, NULL);  
    return 0;  
}
```